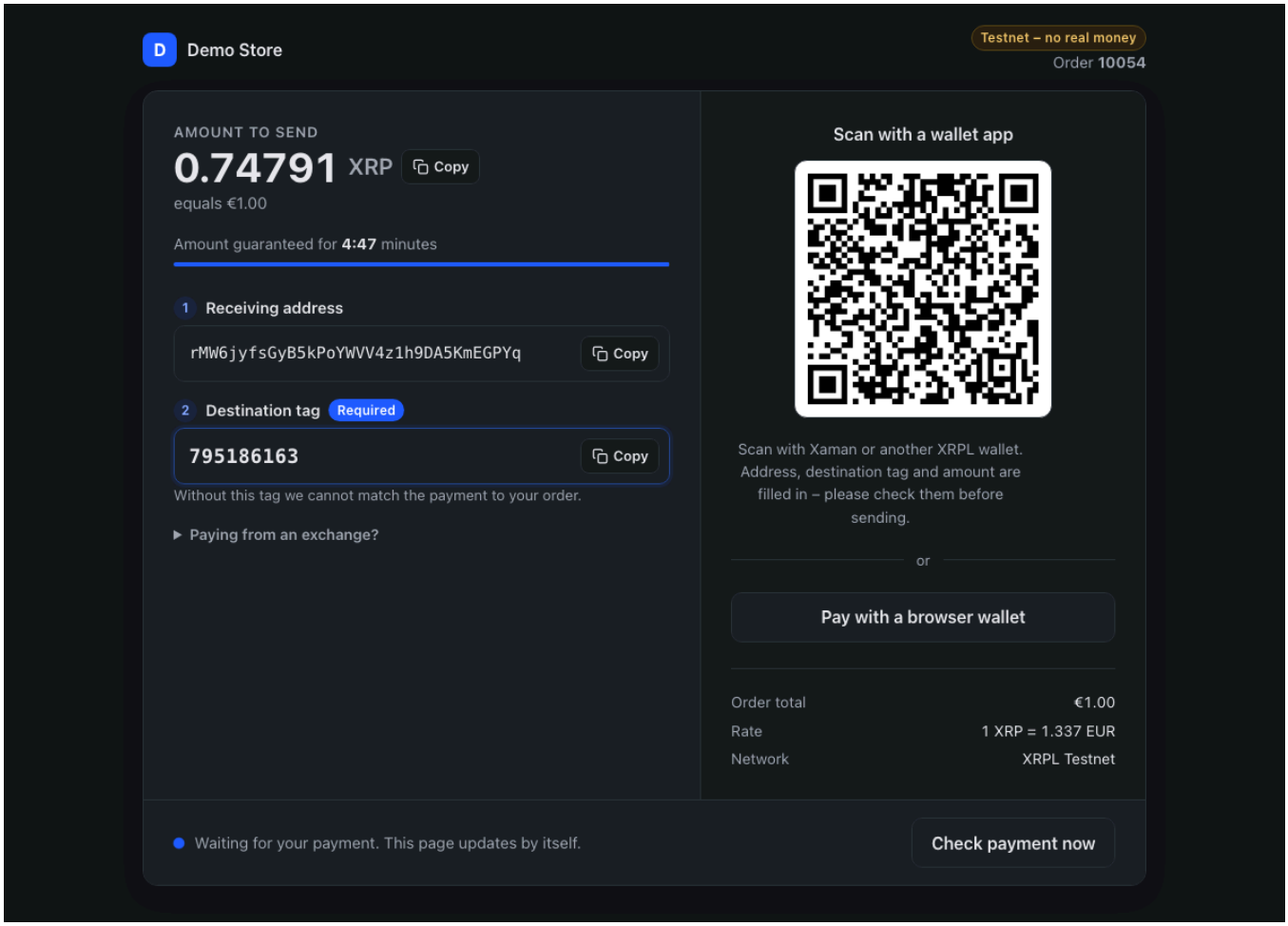
LedgerDirect for Magento 2



Accept XRP, RLUSD and USDC directly on the XRP Ledger, and XAH on Xahau — no payment processor, no custody, funds land in the merchant's own wallet.

The module is the Magento 2 adapter over `hardcastle/ledger-direct-core`, the shared package that holds the XRPL and pricing logic. Everything platform-specific — the payment methods, order statuses, invoices and emails, the payment page, the configuration, the cron job — lives here; price conversion, the oracle set, the stablecoin registry and ledger sync live in the core and are not reimplemented.

Project website: https://www.ledger-direct.com



Try it in five minutes

A complete Magento shop with the module, on the XRP Ledger and Xahau **testnets** — no Magento account, no access keys, no real money. You need Docker and about 3 GB of disk.

```
git clone https://github.com/ledger-direct/ledger-direct-magento2.git
cd ledger-direct-magento2/dev/try
docker compose up -d --wait        # first start installs the shop: about a minute
```

1. Open http://localhost:8088/: the home page explains the demo and offers three hand-painted **miniatures** (2–4 EUR). Put one in the cart and check out as a guest with *Pay with XRP*, *RLUSD*, *USDC* or *XAH*. Their pictures and texts are not under the MIT licence; see `dev/try/catalog/NOTICE.md`.
2. The payment page shows the amount, the receiving account and the destination tag. Pay it from a testnet wallet — Xaman in testnet mode or Crossmark — funded from the XRP Testnet faucet; RLUSD and USDC come from the RLUSD faucet and the Circle faucet, XAH from the Xahau testnet faucet (Xaman on the Xahau testnet). Send the amount exactly as shown, **with the destination tag**, or connect the wallet with the buttons on the page.
3. The page notices the payment within seconds and moves on to the order; the order is invoiced and *Processing* in the admin at http://localhost:8088/admin (demo login `demo` / `demo-ledgerdirect-1` — a local demo login, never reuse it).

On its first start the shop creates its own testnet wallet from the faucet and sets the trust lines to the RLUSD and USDC issuers on it, so all three assets can be paid right away (`docker compose logs phpfpm` names the address). For XAH it creates a Xahau testnet account of its own and has the Xahau faucet fund it: one account per chain, since the same address is a different account on each. The wallets' seeds stay in the stack's volume, testnet only. When the testnet has been reset, the next start notices and creates a new wallet. To receive into an address of your own instead — it then needs the trust lines itself:

```
LD_TESTNET_ACCOUNT=r... docker compose up -d --wait
```

Setting `LD_TESTNET_ACCOUNT` later works too: the next `up` applies it to the existing shop. It sets the XRP Ledger account only; XAH keeps going to the shop's own Xahau account, and every start sets it again. If the Xahau testnet or its faucet does not answer, the XAH settings stay as they were (on a fresh shop: XAH off) and the next `up` tries again. `LD_TRY_PORT` moves the shop off port 8088. `docker compose down` stops the stack and keeps the shop and its orders; `docker compose down -v` resets everything, and the next `up` installs a fresh shop. `docker compose up --build` builds the image from your checkout instead of pulling the published one. Magento Open Source comes from Mage-OS's public mirror; it is not an official Magento image, and the shop is for trying the module, not for production.

How it works

A customer picks XRP, RLUSD or USDC at checkout; each is a payment method of its own. The order is placed straight away as **Pending Payment** and the customer gets a payment page: the exact amount, the shop's receiving address, and a **destination tag** unique to that order. The tag is what ties an incoming ledger transaction back to the order, so one wallet address serves every customer. Right

after the order is placed the customer also receives an email with the link to that page (on by default, see Configuration), so a closed tab is not the end of the payment; the order confirmation itself is held back until the payment has arrived.

The quote is fixed for a configurable window (five minutes by default). Reloading the page never changes the amount; once the quote lapses the customer can ask for an updated one, and the destination tag stays the same so a payment already in flight is not orphaned.

Payments are confirmed three ways, which is deliberate redundancy:

- the payment page polls while the customer is watching,
- a **check now** button for anyone who would rather not wait (and for browsers without JavaScript),
- a cron job that settles orders for customers who closed the page.

An order is credited only from the ledger's `delivered_amount`, only when what arrived covers the requested amount (XRP within 0.15 %, a stablecoin at least the quoted value), and — for stablecoins — only when the currency and issuer match: the core's `SettlementPolicy`. Several payments in the quoted asset add up, so a customer who sent too little can send the rest. Anything else stays open, and the payment page says why: it shows one of five states — waiting, expired, partial payment (with the outstanding amount), wrong token (with the full amount still due), or paid — and updates in place while the customer watches, without reloading.

Once a payment settles the order, the module creates an offline-captured invoice, moves the order to the method's *settled status* (`processing` by default) and sends the order and invoice emails. A payment that arrives but does not pay the order moves it to its own status, **XRPL payment incomplete** (state `pending_payment`), with the amounts and the transaction hash in the status history; the order keeps being matched, and a top-up of the shortfall settles it. Magento's own job that cancels orders left in `pending_payment` after *Pending Payment Order Lifetime* (480 minutes by default) does not touch an order with that status, because there is real money on the ledger for it. The order's *Payment Information* — on the admin order view, in the customer's order view and in the order emails — lists what was quoted (asset, amount, rate, receiving account, destination tag, issuer for a stablecoin), the payment status, what arrived and what is still due, and the transaction hash as a link to the chain's explorer.

The page asks the server every 8 seconds. That endpoint syncs with the XRPL node at most once every 5 seconds per receiving account, whatever the number of customers waiting; in between it answers from what is already stored. A guest order can poll too — the link carries the order's key (Magento's own `protect_code`, the secret behind the guest order view), and no login is required.

Requirements

- Magento Open Source or Adobe Commerce 2.4.7, 2.4.8 or 2.4.9
- PHP 8.2, 8.3, 8.4 or 8.5, as far as the Magento version allows it
- A running Magento cron, for the settlement job

- An XRP Ledger account. Accepting RLUSD or USDC additionally requires a **trustline** to the respective issuer on that account, or payments will fail on the ledger.
- For XAH, an account on Xahau. Use one that receives on Xahau alone: the same address is a different account on each chain, and the module keeps the two apart.

Installation

The module is installed with Composer from Packagist; the git tags are the versions. Run the commands in the root directory of the Magento installation, the one with Magento's own `composer.json`:

```
composer require hardcastle/ledger-direct-magento2
bin/magento setup:upgrade
bin/magento setup:di:compile
bin/magento setup:static-content:deploy -f
```

The module depends on `magento/framework`, which Composer only finds through `repo.magento.com` as configured in that project. Run in an empty directory, `composer require` therefore reports that `magento/framework` "could not be found in any version"; that is expected for any Magento extension and not a problem with the package. Composer 2.9 and later also refuses to resolve dependencies that have known security advisories, and older Magento patch releases lock several of them — if `composer require` fails with "affected by security advisories", update Magento to its current patch release first.

This also installs `hardcastle/ledger-direct-core` from Packagist. If you place the module under `app/code/Hardcastle/LedgerDirect` instead, Composer does not read the module's own `composer.json`; require its two dependencies at project level yourself (`composer require hardcastle/ledger-direct-core:^0.11.2 "bacon/bacon-qr-code:^2.0 || ^3.0"`, the QR-code library renders the payment page's code for browsers without JavaScript) and run the same three `bin/magento` commands.

Installing creates two tables for ledger data (synced transactions and the destination-tag counter) and the order status *XRPL payment incomplete*. Updates run through `setup:upgrade` as usual.

`bin/magento module:uninstall Hardcastle_LedgerDirect` keeps the module's tables. Adding `--remove-data` drops them, including the destination-tag counter — the only record of which tags were already issued. A reinstall then starts a fresh counter, and payments for orders that are still waiting can no longer be matched. Only remove the data once no LedgerDirect order is open.

Configuration

Stores → Configuration → Sales → Payment Methods → Ledger Direct / XRPL:

Setting	
Use testnet	The receiving account must belong to the selected network.
Account, testnet and mainnet	The shop's receiving account on each network; the one for the selected network is used.
Quote validity (seconds)	How long a quoted amount stays fixed. Default 300.
Send payment instructions email, sender, template	The email with the link to the payment page, sent right after the order is placed. On by default.
Payment page: logo	What the payment page shows in its header: the store logo, an uploaded picture (PNG, JPG, SVG or WebP, at most 160 by 32 pixels), or the first letter of the store name.
Payment page: accent colour	The one colour of the payment page — buttons, countdown bar, destination tag — with white text on it, so it must be dark enough (contrast 4.5:1). Default `#1f5eff`; a lighter colour is refused on save.
Payment page: Xaman API key, WalletConnect project id	Public identifiers, optional. With one of them, customers on a phone get an "Open in wallet app" button. Never the API secret.

Each asset is a payment method of its own, with its *Enabled* switch and the title the checkout shows, in the groups *Ledger Direct – XRP (XRPL)*, *Ledger Direct – RLUSD (XRPL)* and *Ledger Direct – USDC (XRPL)* next to it. XRP is on by default; the stablecoins are off by default and need the trustline on the receiving account first.

Issuer addresses are not configurable. They are fixed in the core: a wrong issuer would send customer funds to a dead trustline.

Xahau has a group of its own, *Ledger Direct / Xahau*, with *Use testnet* and the receiving account per network, and the payment method *Ledger Direct – XAH (Xahau)* next to it, off by default. Xahau speaks the XRP Ledger's protocol on a network of its own: payments in XAH by destination tag, priced from Coingecko and Bitrue (XAH trades on Bitrue alone; a euro price rests on Coingecko). The payment request behind the QR code names the Xahau network, so Xaman offers to switch if it is set to another one; the browser-wallet buttons stay off on a Xahau page. Xahau's transactions and tag counter live in tables of their own (`ledger_direct_xahau_tx`, `ledger_direct_xahau_destination_tag`), shared with the XRP Ledger's in nothing.

The cron job `ledger_direct_settle_pending_orders` runs every five minutes with Magento's cron and makes one node request per receiving account and network, then matches every open order against the stored transactions. Without it, a customer who closes the payment page before their transaction confirms is only settled the next time someone visits their payment page.

Test payments

Switch *Use testnet* on and enter a testnet account. Test accounts come from the XRP Testnet faucet for XRP, the RLUSD faucet for RLUSD and the Circle faucet for USDC; the receiving account needs the trustlines, the paying wallet needs the tokens.

The payment page

After checkout the customer lands on `/ledger-direct/payment/index?id=<order entity id>&key=<protect_code>`, a page on an empty layout — no header, footer or navigation — with the store's logo or monogram in its header. A wrong key or id is refused with 403 without saying whether the order exists. The page polls `/ledger-direct/payment/status` with the same parameters, which returns the same payload as every other LedgerDirect plugin (`INVARIANTS.md` in the core, "Payment status"), plus a `redirect` URL once the order no longer waits for payment — whether it was paid on the ledger or canceled by the merchant.

Behaviour and design come from `@ledger-direct/payment-ui`, the package every LedgerDirect plugin shares: the five payment states, the countdown, polling, copy buttons, the QR code with address, destination tag and amount, and browser wallets over XRPL Connect (Crossmark, GemWallet, MetaMask Snap, Ledger, Otsu, Xyra; Xaman and WalletConnect with the identifiers above). The module has no build step: it ships the package's built files under `view/frontend/web/`.

File	From the package	Loaded
`view/frontend/web/css/payment-page.min.css`	`dist/payment-page.css`	with the page
`view/frontend/web/js/ledger-direct-payment-ui/payment-page.min.js`	`dist/payment-page.js`	with the page
`view/frontend/web/js/ledger-direct-payment-ui/wallets.min.js`	`dist/wallets.js`	only when the customer opens the wallet list, by a native `import()`

The files carry `.min.` in their name so Magento's own minifier leaves them alone. `view/frontend/web/js/ledger-direct-payment-ui/VERSION` names the package tag the files were copied from. To update: copy the three files from the package's `dist/` at the new tag to those names and write the tag into `VERSION`. The template renders the package's markup contract (`src/README.md` there) from a view model; every sentence a customer reads is in `view/frontend/templates/payment/index.phtml` and `i18n/*.csv`, nothing is rounded or reformatted in the browser.

Without JavaScript the page still works: the amount, address and tag are server-rendered, the QR code is drawn on the server from the same payment request, and the "Check payment now" button is a plain form that reloads the page, which syncs and settles.

External services

Exchange rates come from the public APIs of Coingecko, Binance and Kraken, through the core. No personal or payment data is sent to them; only the current rate is requested when a payment is quoted or displayed, and rates are cached briefly in Magento's cache so a short outage of one source does not interrupt checkout.

- Coingecko API: Terms of Service, Privacy Policy
- Binance API: Terms of Use, Privacy Policy
- Kraken API: Terms of Service, Privacy Policy

Development

How the whole of LedgerDirect is tested across the core, the shared page package and the four plugins — the layers, what each catches, the nightly end-to-end runs and the manual cases — is in `docs/testing.md` of the core. The manual cases for this module, by the core's case IDs, are in `docs/manual-tests/payment-status.md`.

The module is developed inside a Magento project, under `app/code/Hardcastle/LedgerDirect`, whose Composer autoloader the tests use. From the Magento root:

```
vendor/bin/phpunit -c app/code/Hardcastle/LedgerDirect/tests/phpunit.xml.dist
```

The unit suite runs the real core services and stubs only the edges this module owns: the HTTP client, the transaction repository port and Magento's repositories. It is offline by design and needs no database. `tests/README.md` has the details, including the Magento Marketplace keys the weekly cross-check against `repo.magento.com` uses (no other job needs them).

To work against a local core checkout instead of the released version, add a path repository to the *Magento project's* `composer.json` and require the branch; keep the module's own constraint on the released version:

```
composer config repositories.ledger-direct-core path ../LedgerDirectCorePHP/ledger-direct-core-php
composer require hardcastle/ledger-direct-core:dev-master
```

Continuous integration and release

Every push and pull request runs `.github/workflows/ci.yml`: PHP syntax, the Magento 2 coding standard (`magento/magento-coding-standard`), and the PHPUnit suite inside a real Magento 2.4.7, 2.4.8 and 2.4.9 project, each on the newest PHP it supports (8.3, 8.4, 8.5). Magento Open Source comes from Mage-OS's public mirror, which needs no access keys — pull requests from forks are tested too — and is several times faster than `repo.magento.com`; Composer's package cache is kept

between runs. Once a week, and on demand, the same tests run against `repo.magento.com` itself (`MAGENTO_COMPOSER_AUTH` secret with Marketplace access keys; without it that job logs a warning and skips), so a gap in the mirror shows up. A pull request that changes only documentation (Markdown, `docs/`, the changelog and the like) runs PHP syntax alone; when the job that classifies the changes fails, everything runs.

`.github/workflows/try.yml` builds the try stack (`dev/try`) from the checkout, starts it and runs `dev/try/smoke.sh` against it, then walks the storefront in a real browser. A pull request skips it when it changes only documentation, tests, test tool settings or other workflows. `.github/workflows/e2e.yml` drives the core's case catalogue against that try stack with the harness `ledger-direct-e2e`, first on the XRPL testnet, then on the Xahau testnet (where the harness switches Xahau on in the shop and PS-04 is skipped, Xahau having no issued asset): on a version tag (every case but the 35-minute wait), every Monday at 05:17 UTC (every case on the XRPL testnet, on Xahau every case but that wait) and by hand. One CI treasury per chain, from the repository secrets (`LEDGERDIRECT_CI_XRPL_TREASURY_*`, `LEDGERDIRECT_CI_XAHAU_TREASURY_*`). It does not gate a merge: the testnets and their faucets are not deterministic.

`ci.yml` and `try.yml` each end in one job that sums up the others, *CI passed* and *Try passed*. These two are the checks a merge into `main` requires. A job skipped on purpose (a pull request that changes only documentation) counts as passed, a failed or cancelled one does not.

The rules for `main` are a GitHub ruleset, kept in `.github/rulesets/main.json`: changes only through a pull request, no approval needed, *CI passed* and *Try passed* green on its last commit, merged as a squash or a merge commit; no force push, no deletion. Repository admins may merge a pull request without the checks, never push to `main` directly. A change to the rules is a pull request that changes the file; after the merge it is applied on GitHub (`gh api -X PUT repos/<owner>/<repo>/rulesets/<id> --input .github/rulesets/main.json`, the first time `-X POST repos/<owner>/<repo>/rulesets`). Every Monday, and on demand, `ci.yml` checks that the ruleset on GitHub matches the file (`.github/rulesets/check.sh`). Who may bypass the rules is not part of that check: GitHub shows the bypass list only to admins of the repository.

Tags have a ruleset of their own, `.github/rulesets/tags.json`: a tag can be created, never moved or deleted, so a version on Packagist always stands for the same commit. It is applied and checked the same way.

A release is a git tag, which Packagist picks up and Composer resolves as the module's version; `composer.json` carries no version field of its own. `CHANGELOG.md` lists the versions.

The Adobe Commerce Marketplace wants the same code as a zip with `version` in its `composer.json`: `bin/marketplace-zip.sh <tag>` builds `build/Hardcastle_LedgerDirect-<tag>.zip` from the tag, writes the version into the copy, checks the fields the technical review requires and leaves tests and tooling out (`.gitattributes` export-ignore, the same rules Packagist applies). The Marketplace takes only packages under the seller's vendor name, so the copy is named `ledger-direct/ledger-direct-magento2` and replaces `hardcastle/ledger-direct-magento2`; module name and namespace stay the

same. The listing's assets live in `docs/marketplace/`: the icon, the user guide (`bin/docs-pdf.sh` renders it from this README) and the release notes per version.

Translations

English and German, as `i18n/en_US.csv` and `i18n/de_DE.csv`.

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